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ORIGINAL ARTICLE

Leveraging online reviews to decode quality-induced customer dissatisfaction: From perception to product discouragement

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Correction added on July 1, 2026, after first online publication: An affiliation has been added for Kristof Coussement.

Abstract

E-commerce practitioners and researchers recognize that quality concerns are the primary drivers of customer dissatisfaction with products or services. While dissatisfaction can arise from various factors, little is known about quality and its components, specifically from the perspective of dissatisfied customers. Grounded in the foundational principles of expectancy conformance theory and emotional regulation theory, our study investigates the key characteristics driving quality-induced customer dissatisfaction and their influence on consumers' response behaviors. We further examine how ways of expressions and feelings underlying reviews nudge future recommendations. By combining natural language processing and statistical modeling for around a million online reviews, we uncover and identify the characteristics underlying the sources of quality-induced customer dissatisfaction. Our findings highlight the intermediary role of negative sentiments and emotions, shifting the focus from regular defects or design-related stand-alone issues for the practice. Rather, it is the customers' affective states, escalating from mild dissatisfaction to strong frustration, which mediate the impact on future recommendations and can lead to extreme reactions such as product discouragement. Therefore, portal managers can apply our findings to enhance decision-making in complex situations by developing coping strategies to regulate affective states of disappointed customers and thereby curb negative word-of-mouth.

KEYWORDS

artificial intelligence (AI), business analytics, customer dissatisfaction, e-commerce, natural language processing (NLP), product discouragement, quality

1 | Introduction

Customer satisfaction and grievance redressal mechanisms are critical enablers for organization's ability to retain customers (Koufteros et al., 2014; Tsikriktsis & Heineke, 2004). For firms and especially the e-commerce, customer grievances and dissatisfaction can cause customers to exit from future transactions (Tirunillai & Tellis, 2014). For online portals, dissatisfied customers can voice their concerns about poor quality and negative experiences, which influences the purchase decisions of a large chunk of potential customers (Ferguson & Johnston, 2011), resulting in reduced sales and loss of competitive advantage (Chatterjee, 2019). Thus, addressing dissatisfied customers' quality-related concerns is crucial in ensuring a loyal customer base for the e-portals. Despite the importance of customer satisfaction, limited works have attempted to unveil the antecedents and consequences of customer dissatisfaction, from a quality standpoint. Previous works acknowledge that dissatisfaction stemming from quality issues is hard to understand and that identifying the source(s) of disappointment is challenging (Lapré, 2011; Vana & Lambrecht, 2021). The sheer volume

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of transactions and information in online portals makes it even more challenging to trace the quality-induced sources and constituents of customer dissatisfaction (Kim, 2021).

Empirical studies on how large numbers of disappointed customers perceive quality over e-portals, and how such perceptions drive negative response behaviors remain scarce, with little effort made to identify the underlying sources of quality-related grievances (Abrahams et al., 2015; Ma et al., 2021; Vana & Lambrecht, 2021). Existing notions of online quality largely overlook the specific precursors and attributes of customer dissatisfaction, as they rely on traditional methods with limited samples and infrequent assessment (Ahani et al., 2019; Lapré, 2011; Tirunillai & Tellis, 2012). Thus, understanding the concerns underlying quality-induced dissatisfaction of customers remains crucial to pre-empt the future possibilities of firm failure. As a result, we develop a model to empirically examine the characteristics underlying the root causes of quality-induced source of dissatisfaction and demonstrate how these latent origins shape customer response behaviors.

1.1 | Background and motivations

This study is mainly motivated by the rich discussions we had with portal managers, e-commerce consultants, and other practitioners of the platform ecosystem. The interactions originate from an interesting academia–industry engagement, in which the institute of affiliation of the first author signed a memorandum of understanding to onboard artisans on Flipkart.com. Thereafter, several interactions took place with their chief corporate affairs officer and director, east region, Flipkart group. Their top brass shared that customer satisfaction is the key to the success of e-commerce. Accordingly, their research team ensures that customer satisfaction is the top priority and channels their effort to identify the sources of customer dissatisfaction. Later, discussions helped us understand the broad reasons for customer dissatisfaction. The practitioners acknowledged that identifying the sources of dissatisfaction, like product defects or inaccuracies, is crucial, and addressing these issues requires implementing robust quality control measures throughout their supply chain and network agents.

We started engaging with other entities of the platform ecosystem: technology providers, procurement veterans, product manufacturers, and the chief of materials (see the Appendix in the Supporting Information). Due to high return rates, last-mile delivery is still challenging (Lu et al., 2024). Practitioners had severe reservations about product quality and return rates, which led to increased costs and diminished profitability. E-commerce managers say that proactive quality management strategies can mitigate these challenges. The practice could minimize return incidences and improve financial performance by detecting and resolving quality issues early in the product lifecycle. The industry leaders talked about customer dissatisfaction arising from quality issues and how this could tarnish their brand through negative reviews and word of mouth. To safeguard their reputation, it is essential to promptly address the quality-related concerns raised by customers. Quality-related problems constitute most of the customer discontent and resentment toward any product or service (Abrahams et al., 2015; Ferguson & Johnston, 2011).

So far, quality has been envisaged from the customer satisfaction standpoint (Ferguson & Johnston, 2011; Wells et al., 2011; Zaman et al., 2021), not from the standpoint of customer dissatisfaction. Furthermore, the quality attributes of e-commerce, an information-intensive ecosystem, have been primarily technical, interface, or website-related and the available instruments of overall quality are restricted to certain product categories (He et al., 2018), sectors (Zaman et al., 2021), or industries (Kim, 2021; Lapré, 2011; Lapré & Tsikriktsis, 2006).

The practitioners expressed concerns over the constraints to the success of e-commerce. Website navigation, payment gateways, freebies, and price wars influence customer engagement and satisfaction, which are perennial concerns[1,2], but product return is the most pressing challenge.[3,4] Returns due to quality issues—mainly attributed to damaged, defective, or perceptually "inferior" products—pose a ripple effect on sporadic inventory, raising shipping costs and redress expenses. When raising return requests, the platform engagement experience is unpleasant (Schroeder et al., 2005). However, given the hierarchical nature of quality perception—several layers of characteristics underlie the consumer psyche—the task is yet to be achieved (Moody, 2005). This warrants the need to focus on a process initiative while tackling design and/or defect-related performance issues, premature failure owing to below-par material, or failed service-level guarantees (Moody, 2005; Schroeder et al., 2005).

Based on our reasoning, we believe that design flaws, defects, material concerns, and service failures all have a detrimental impact on customer satisfaction, trust, and branding. Poor design can result in usability issues, annoyance, and decreased product value. Defects and quality difficulties reduce trust and cause unfavorable word-of-mouth. Poor materials lead to poor durability and early product failures. Finally, bad customer service can escalate minor issues into big discontent, resulting in loss of repeat business. Therefore, e-portal managers are keen on tackling the problem and grievance redressal strategies for issues pivoting around poor performance and premature failures owing to defect, design, material, and service.r

The notions of perceived online quality do not describe the precursors and attributes of customer dissatisfaction (Ahani et al., 2019; Lapré, 2011). Even the existing multidimensional notions of quality rely on conventional qualitative or mixed method approaches, based on surveys and interviews, using a limited sample of consumers, lacking periodic assessment

[1] https://vue.ai/blog/ai-in-retail/ecommerce-challenges-in-2021/.
[2] https://magazine.insightech.com/the-most-common-website-bugs-and-errors-in-ecommerce/.
[3] https://www.mageants.com/blog/customer-service-issues-in-ecommerce.html.
[4] https://www.flipkart.com/pages/ASC-policy-store-1.

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and do not allow for an appreciation of online quality (Ahani et al., 2019; Archer & Wesolowsky, 1996; Tirunillai & Tellis, 2012, 2014).

Only a few have studied the quality-stimulated customer dissatisfaction for e-commerce channels (Abrahams et al., 2015; Ma et al., 2021). Empirical research into how a large mass of disappointed customers perceive online quality and how their perception influences negative response behaviors is limited (Ferguson & Johnston, 2011; Lapré, 2011; Vana & Lambrecht, 2021; Wells et al., 2011). No attempt has been made to identify the sources or constituents of quality concerns pertaining to customer dissatisfaction. Two challenges limit the research into quality-related grievances.

First, millions of customers post reviews evaluating their experiences, and this information overload makes it difficult to trace the sources of quality issues (He et al., 2018). Amazon's in-house speaker, Echo Dot, alone garnered more than a million posts within the first few years of its launch (Jabr & Rahman, 2022). Such volume exhausts readers of reviews (Liu et al., 2024); battling noise and irrelevance, the human mind fails to distinguish hidden and true signals (Jabr & Rahman, 2022). This calls for the practice and e-portals to find unique, innovative ways of detecting clear, consistent signals and benefits from reviews.

Second, we do not have the theoretical foundations to uncover customer perception and the responses to quality-induced sources of customer dissatisfaction (QISCD; Ma et al., 2013; Vana & Lambrecht, 2021). Customer resentment can be attributed to various sources, but several issues might underlie product or service quality discontent (He et al., 2018). For example, a review may mention an erratic noise, but that is not enough to diagnose the root cause because whether that noise is due to a defect in a unit or a design issue that affects the entire production line remains unclear. Likewise, scarcity of materials or poor responsiveness of customer support may make parts or the product unavailable. Dissatisfaction can have many sources, and these sources differentially influence the nature and extent of dissatisfaction. Firms must know the dimensions of the sources of dissatisfaction since these influence consumer experience and response behavior (Tirunillai & Tellis, 2014). Our conceptualization is based on the underpinnings of the expectancy conformance theory (ECT; Hsu & Lin, 2015) and the theory of emotional regulation (ER; Gross, 2002).

The literature on consumer behavior is motivated largely by persuasion theories, especially the expectancy (dis) confirmation paradigm. ECT explains how prior expectations shape customer evaluation of product (or service) performance and how dissatisfaction occurs when the formed expectations are unmet (Hsu et al., 2012). In the realm of ECT, this is negative disconfirmation, as shown in the third block of the above framework diagram in Figure 1. These unmet expectations then cause negative sentiments and emotions like anger or frustration. The theory of ER finally explains how individuals regulate their emotions and their subsequent influence on negative response behaviors (Lin et al., 2009). These behaviors could be actions like posting negative reviews, voicing complaints, discouraging others from buying and switching between brands (Kuo & Wu, 2012; Lin et al., 2009). By complementing ECT and ER, we can uncover the complete pathway of unmet customer expectations that leads to emotional reactions, which then manifest into customer behaviors.

Summarily, the confluence of ECT and the theory of ER helps in understanding both the emergence and expressions of customer dissatisfaction and their impact on negative response behaviors. Based on these arguments, theoretical foundations, and literature synthesis, we conclude that there are no promising attempts to mine the consumer perception of the features of QISCD and their impact on customer responses. So far, the research into customer satisfaction has ignored the precursors of customer dissatisfaction (Hult et al., 2019). Until recently, online reviews have been used to promote products, loyalty, and sales; however, no work has explored the review content to analyze the sources of resentment (Jabr & Rahman, 2022; Wang et al., 2024). Neither the configurations of the sources of customer dissatisfaction nor their impact can be easily diagnosed. This dearth of insight(s) is pronounced for mediums such as e-commerce, which experience an overwhelming influx of customer reviews. Resultantly, our inquiry raises the following research questions:

1. What are the root causes and constituents of QISCD in e-commerce?
2. How does QISCD affect the response behaviors of online consumers?

1.2 | Research overview and intended contributions

To answer these questions, we utilize secondary data from over 0.6 million online customer reviews from Amazon.com. We filter reviews of dissatisfied customers out of the corpus. Using manual analysis of the review summary, we could label and segregate reviews into four predominant QISCD: design, defect, service, and material. Next, we use text mining and natural language processing (NLP) to extract the underlying dimensions of sources of quality concerns. Later, through econometric investigation, we gauge the impact of these dimensions on future recommendations, perceived as one of the most crucial ingredients for online portals, because word of mouth and the opinions of other customers influence market share (Kumar et al., 2023a).

Our results demonstrate that the constituents of QISCD stimulate negative sentiments and emotions, which, in turn, influence negative recommendations. Our key findings underscore the pivotal intermediary role of affective states, moving the discussion beyond isolated sources of dissatisfaction. It is not merely the operational issue itself but the customer's affective response, ranging from mild dissatisfaction to intense frustration, that determines how the experience shapes future recommendations. These escalating emotional states can amplify dissatisfaction, ultimately leading to stronger

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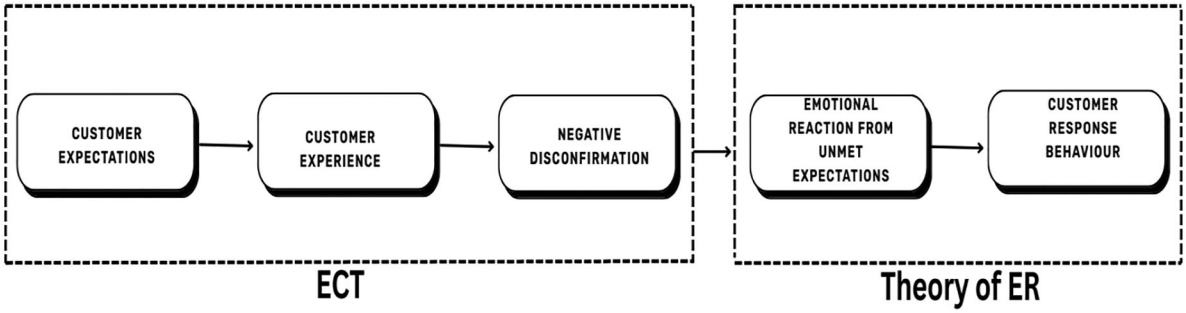


FIGURE 1 Integrated framework diagram depicting the complementarity of expectancy confirmation theory (ECT) and theory of emotional regulation (ER).

outcomes such as discouraging others from purchasing the product. Our results suggest that recovery strategies should extend beyond resolving operational failures to explicitly addressing customers' affective responses, as ER is a critical determinant of post-experience advocacy. Our study makes three unique contributions.

We attempt a deep dive into the sources and constituents of quality concerns from the viewpoint of disappointed customers. Through our comprehensive approach, product owners and portal managers can now gain incremental knowledge about the latent factors that shape the QISCD.

Together, the theories of ER and expectancy conformance provide a foundation for understanding the cognitive factors of consumer decision-making, especially the role of consumer sentiments, emotions, and writing style.

We develop a model to empirically investigate the root causes of negative customer behaviors associated with each quality-induced source of customer dissatisfaction and how its latent source influences future purchases through product discouragement.

The current study will thus assist e-commerce companies with an approach to uncovering the triggering factors behind these sources, proactively addressing customer grievances, and attaining the goal of customer retention. The rest of the paper has the following structure. Section 2 earmarks the theoretical background. Section 3 describes the methodology. Section 4 reports the results. Section 5 discusses the most significant results. Section 6 discusses the theoretical and managerial implications. Section 7 presents the conclusion, limitations, and future directions.

2 | Literature review and hypothesis development

2.1 | Theoretical background

The beginning point for any transaction happens at the pre-consumption level, with customers building perception about the products (or services) they ought to consume. Once the consumption happens, the customers start evaluating their experiences with the expectations. This is where ECT offers a lens to gauge how much the actual performance meets or falls short of the pre-consumption expectations (Liao et al., 2017). The most common outcomes of meeting expectations or confirmation are customer satisfaction, while disconfirmation leads to disappointment (Hsu & Lin, 2015). In reality, the impact of dissatisfied customers could be far reaching, and it becomes imperative to understand the response mechanism after unmet expectations or disconfirm. The theory of ER complements our understanding of the psyche underlying customer behaviors. Once dissatisfaction sets in (from ECT), emotions are dysregulated in terms of sadness, anger, or frustration, leaving customers disappointed (Ebrahim et al., 2016). The ER theory specifically explains how consumers navigate these negative emotions and how they manifest into response behaviors like abandoning products, switching brands, or discouraging others to use.

We tend to summarize the integration of ECT and the theory of ER in four broad stages. ECT remains the cornerstone for (Stage 1) pre-consumption expectations and post-consumption evaluation (Stage 2) resulting in (dis)confirmation. In the third step, the theory of ER then explains how the disconfirmation triggers (Stage 3) negative emotions. Finally, ER helps us to understand how negative emotions lead to negative response behaviors of customers (Stage 4). The below framework diagram (figure 1) depicts the integration of ECT and ER.

2.2 | Hypothesis development

In practice, individual consumers may have different perceptions about product features, components, or attributes (Archer & Wesolowsky, 1996; Ma et al., 2013). The dimensions of product quality are performance, features, conformance, durability, serviceability, aesthetics, and perceived quality (Garvin, 1987). The problems relating to all the seven dimensions are challenging to address simultaneously. However, segmentation can identify the critical dimensions and lead to customer satisfaction and positive future recommendations. E-commerce organizations align their attention toward customer management by focusing on customer ratings and reviews, but only a few studies analyze unstructured text to study issues related to product quality (Abrahams et al., 2015), possibly because product and service quality are multidimensional constructs and measuring them is difficult. We present a table summarizing the seminal works which

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are related with idea of customer decision-making based on quality evaluations. Table 1

From a customer viewpoint, quality is a blend of product and service superiority. Historically, quality has also been defined as relating to customer satisfaction. Customer reviews are instrumental in shaping the buying behavior of other customers as they trust and believe such reviews (Benlian et al., 2012). These reviews have the potential to directly influence future recommendations (Chatterjee, 2019). Thus, quality-related issues that affect customers are an amalgamation of product and service quality (Zaman et al., 2021). Therefore, we hypothesize that the major QISCD will influence customer evaluations and word-of-mouth and that:

Hypothesis 1. *The four QISCD—design, defect, service, material—affect future recommendations on e-portals*.

Efficiently extracting product defect data from online customer reviews is a non-trivial task with vast utility for businesses (Wang et al., 2018). Law et al. (2017) use text-based classification to identify significant terms about malfunctioning of automobile components. Winkler et al. (2016) use text mining to identify characteristics of toys that might harm children; the study compares millions of Amazon reviews to a smoke list based on interviews to identify toy defects as described in customer reviews. Zhang et al. (2019) use a latent Dirichlet allocation topic model to identify the types of defects as described in user complaints. Therefore, identifying defects through online customer reviews is imperative, and we hypothesize that defects and their underlying aspects will impact customer response behaviors.

Hypothesis 2. *Defects and their underlying factors influence future recommendations on e-portals*.

Online customer feedback shows that design is a critical dimension of product quality. Product design is market-driven; to understand the user experience, the practice considers user preferences for design and content (Yang et al., 2019). Ji et al. (2014) prioritize the user concerns raised in online reviews and integrate them into product design. Studying online reviews has several advantages, but only a few analyze online studies to understand user feedback on design (Yang et al., 2019). Survey-based qualitative studies based on user feedback have been undertaken to improve product design (Hassenzahl, 2010). Thus, design is a crucial component of customer feedback for pre-empting quality concerns, and hidden design issues influence future response behaviors.

Hypothesis 3. *Design and its underlying factors influence future recommendations on e-portals*.

The quality of the packaging and material affects consumer satisfaction and loyalty as customers raise issues with it. Therefore, material and packaging are ways to provide consumers with safe products and market the company (Brody, 2002). Still, few studies incorporate online customer feedback—almost all the studies are based on survey questionnaires. This article bridges the gap by using online customer complaints against the material used in products.

Hypothesis 4. *Material and its underlying factors influence future recommendations on e-portals*.

Customers assess service quality based on whether their needs are met. Reading online reviews provides insights into service quality without consuming an organization's services. On online e-commerce platforms, service quality is a critical dimension of customer satisfaction because switching cost is low (Wen et al., 2014), and maintaining service quality (replacement of defective items, delivery time, customer care, etc.) is therefore a critical aspect of quality management. The literature on service failure recognizes that what is observed is the tip; it is the need of the hour to uncover the underlying or latent attributes of service quality.

Hypothesis 5. *Service and its underlying factors influence future recommendations on e-portals*.

Based on the theoretical foundations of ECT and ER theory, we postulate that quality characteristics influence consumer evaluation of performance and spur emotional cognizance and polarized sentiments (Gross, 2002). The theory of ER posits that such evaluation should ultimately result in customer response behaviors, while ECT states that emotional cognizance plays an intermediary role (Section 2.1) Sentiments and emotional valence influence buying intention and subsequent customer behaviors. Bougie et al. (2003) show that poor service may trigger anger and disgust and shape dissatisfaction and negative response behaviors. Likewise, but more critical, is the negative experience from poor product performance, as frustration may flare into aggression, resulting in highly negative word of mouth and impeding future purchases. Sentiments and emotional cognizance are precursors of customer satisfaction and future response behavior, mainly when it comes to spreading pleasant or unhappy experiences through word of mouth (Chatterjee, 2019). Thus, it becomes imperative to analyze how much QISCD and their latent dimensions spur emotions capable of controlling response behaviors. Resultantly, we propose the following hypotheses regarding the intermediary role of emotions and sentiments in the overall consumer evaluation and decision-making journey:

Hypothesis 6. *Emotions mediate the relationship between the source(s) of quality-induced customer dissatisfaction and future recommendation for online portals*.

Hypothesis 7. *Sentiments mediate the relationship between the source(s) of quality-induced customer dissatisfaction and future recommendation for online portals*.

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TABLE 1 Contributions of seminal and relevant articles.

	Contributions			
Source	**Notion of quality**	**Constituents of quality**	**Business impact**	**Validation**
Wen et al. (2014)	E-commerce service quality	Website and transaction quality	Customer satisfaction and loyalty	Based on the Engel–Kollat–Blackwell customer's decision-making process, a survey was conducted for 717 online consumers
Zaman et al. (2021)	E-commerce service quality	Service attributes pertinent to healthcare	Patient satisfaction	Utilized text mining on reviews posted on Facebook
Ko et al. (2019)	Quality of healthcare	Technical and interpersonal	Operational efficiency and, ultimately, patient satisfaction	Using text mining, the work explores the technical and interpersonal nuances of quality from a million physician ratings and narratives
Tirunillai and Tellis (2014)	Quality as an aspect of customer satisfaction	Product quality attributes	Brand segmentation	The study performed topic modeling on reviews collected from 15 firms over 4 years
He et al. (2018)	Quality based on experience	Mobile phone product quality	Scale development	Text mining and fuzzy comprehensive evaluation were used to develop a quality index for mobile phones
Lapré (2011)	Prevent customer dissatisfaction from past failures	Service quality-related attributes	Customer dissatisfaction levels	Using the organizational learning curve, the work reported on the quarterly failures of nine US airlines over 11 years
Vana and Lambrecht (2021)	Ratings and reviews reflect overall quality and customer satisfaction	Identified factors influencing sequence of display of reviews	Future purchases	The study measures the impact of review ratings on sequence of display for a 14-day window for e-tailers
Wells et al. (2011)	Quality for online portals	Website and product quality	Purchase intentions	The work was motivated by signaling theory conducted over three controlled experiments
Ferguson and Johnston (2011)	Customer dissatisfaction through quality	The study was silent on the factors leading to customer dissatisfaction	Discussed various dissatisfaction-induced response behaviors	Based on Hirschman's Response Framework, a comprehensive literature review was conducted about dissatisfaction-induced response behaviors
Kim (2021)	Discussed the sanctity of consumer expressions and online reviews	Information cues such as product information, brand reference, and so forth	Purchase likelihood	Using signaling theory, the study captures reviews and other information from a travel website to arrive at their conclusion
Current study	Focuses on understanding quality-induced sources of customer dissatisfaction (QISCD) and their underlying factors	The latent constituents of QISCD are identified alongside linguistic attributes	The work establishes the role of QISCD on future product discouragement over online portals	Foundational on ECT and the elaboration likelihood model; we analyze millions of online reviews to understand signals of purchase discouragement

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Emotional cognizance and consumer sentiments expressed in online reviews may influence the decision-making of other and potential customers. Effectively, word-of-mouth communication is more widespread if sentiments and emotions are noticeable and vivid in descriptions (reviews). The writing style in reviews leaves an impression on the consumer psyche and affects purchase intention (Wang et al., 2022). Some initial efforts gathered evidence of word count to foster sentiments and emotions and influence recommendation behavior (Chatterjee, 2019). In other related e-commerce applications, linguistic features reflect consumer intentions and their overall psyche (Chatterjee et al., 2021). However, writing style and authorial features have been under-researched, particularly for their influence in shaping consumer decisions, intentions, and behaviors (Liu et al., 2020). By writing style and related features, we commonly hint at stop words—articles, prepositions, other pertinent content words conveying contextual meaning, and parts of speech (Bang et al., 2021). Dissatisfaction is more critical regarding the manner of expression—silent or otherwise. Individual traits and characteristics, mostly demographics and personality, influence expression (Bougie et al., 2003), but the statistical evidence is inconclusive. More remarkable is the way of expression, which has remained under-researched in terms of style of expression (writing)—for example, whether the customer is tolerant or confrontational. Such works also showcase that rigor in expression may distinguish between the extent of frustration and customer anguish. We therefore postulate that:

Hypothesis 8. *Authorial features mediate the relationship between the source(s) of quality-induced customer dissatisfaction and future recommendation for online portals.*

Thus, grounded in the prior literature and based on the review summary of more than a million online consumers, we identify defect, design, material, and service as the four primary QISCD. Defect (performance, conformance), design (aesthetics, features), and material (durability, perceived quality) are dimensions of product quality, whereas service encompasses all the service quality-related issues the customers face. Defect, design, and durability issues are well-known aspects of product quality (Abrahams et al., 2015; Ma et al., 2021). But the role of service quality in shaping e-commerce customer responses is unexplored. Therefore, possible linkages can be explored between the acknowledged sources of product and service quality-related sources of customer dissatisfaction and the defined levels of customer expectations that must be met for businesses to succeed.

3 | Research design

To capture the voices of disappointed customers, we resort to the e-tail giant Amazon.com Inc.[5] (Amazon, hereafter). Amazon is the most popular brand in the realm of e-commerce, and it is particularly known for its laser-like focus on customers and operational proficiency (Kumar et al., 2023a). Amazon[6] encourages customers to share their direct experience of products and associated services on a scale of 1–5 to capture explicit reflections of customer satisfaction. Consumers are also encouraged to submit detailed reviews highlighting the pros and cons of their transactions.

3.1 | Data collection and processing

Given that this study focuses on understanding QISCD, we first filter reviews that are poor in satisfaction levels (rating 1 or 2; Kumar & Bala, 2017; Tang & Guo, 2015). Prior research suggests that ratings of either 1 or 2, on a scale of 1–5, denote unmet expectations and customer dissatisfaction majorly due to negative experiences (Chevalier & Mayzlin, 2006). On the contrary, middle-range ratings like "3" on a scale of 1–5 reflect moderate satisfaction levels, while also reflecting neutral and ambivalent perceptions. Both from consumer psychology and general notions with ratings scale of e-commerce, a rating of 3 is considered undecisive and mostly denoted neither satisfied nor dissatisfied[7] While ratings of 1 or 2 surely hint at problems, a rating of 3 is considered as simply fine.[8] From a total of 602,777 reviews, capturing reviews for lower ratings resulted in 80,361 reviews solely pertaining to dissatisfaction, that is, reviews corresponding to ratings of 1 or 2 on a scale of 1–5. We therefore capture the expressions of certain cases of discontent. This dataset includes reviews between May 1996 and October 2018 for appliances only (Ni et al., 2019). Focusing upon single category, such as appliances alone, we could avoid variability from cross-category differences while ensuring consistent characteristics. Further descriptive analysis revealed that many reviews lacked detailed and product-specific explanations, phrases such as "waste product," "poor/bad quality," "worst buy," and so forth, fail to convey attribute-related meaning.

In continuation of our previous analysis of authorial features, we performed a quintile analysis and discovered that the lowest decile comprised reviews with a maximum of eight words. It was observed that reviews containing eight words or fewer often lacked the ability to convey the subtleties associated with QISCD, thereby contributing to noise within the overall word distribution. This finding underscores the importance of sufficient word count in capturing the nuanced expressions necessary for meaningful review analysis. Therefore, following similar guidelines from related works (Chan & Chong, 2017; Zhang et al., 2008), we removed reviews with less than eight words, from the entire textual corpora. Finally, our corpus consisted of 71,908 dissatisfactory reviews conveying considerable expression.

[5] https://www.amazon.com/.

[6] https://www.forbes.com/sites/brittainladd/2018/08/27/these-two-things-are-what-make-amazon-amazon/?sh=7642cd685fd5.

[7] https://authorchristopherdschmitz.wordpress.com/2017/09/06/what-do-those-stars-mean-on-amazon/.

[8] https://rosieamber.wordpress.com/tag/what-do-star-ratings-mean-on-amazon/.

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TABLE 2 A summary of steps to select the number of reviews for econometric validation.

Step 1	Total number of reviews for appliances	602,777 reviews
Step 2	A subset of reviews from Step 1 with dissatisfactory reviews having a rating of less than 3 (on a scale of 5)	80,361 reviews
Step 3	A subset of reviews from Step 2 with reviews substantially expressing dissatisfaction with at least nine words	71,908 reviews
Step 4	A subset of reviews from Step 3, which reflects future recommendation (filtered using regular expressions programming)	1824 reviews

To align with our objective of capturing how dissatisfaction translates into negative actions, we filtered for reviews that explicitly indicated purchase discouragement. Filtering using regular expressions, we could arrive at a targeted analysis of 1824 reviews echoing behavioral outcomes of customer discontent. This also shielded us from drawing conclusions from spurious significance. From a statistical standpoint, overly large samples (like 72,000 in Stage 3) can distort results by producing artificially low *p*-values, leading to misleading hypothesis rejections driven by sample size rather than true effects (Wagenmakers et al., 2015).

We now distributed these reviews to a section of management science scholars specialized in retail and platform business. Moreover, the demographics were suitable due to their reliance on e-commerce for personal consumption. In a span of 3 months, the section of almost 40 scholars construed that most of the reviews mentioned problems related to defect, design, material, or service. Later, the participants were asked to manually explore traces of future recommendations such as "do not buy," "I still recommend," and so forth. Such phrases helped them classify reviews into either explicitly discouraging future purchases or somewhat encouraging future transactions.

Manual labeling—annotation by individuals—is a common practice in the AI-NLP literature (Fredriksson et al., 2022). The mechanism of manual labeling is human-intensive and tedious. Still, it makes maintaining data quality easy and is deemed suitable for small data-labeling processes involving a few thousands, such as ours, if following the guidelines (Mukherjee & Bala, 2017a, 2017b). First, we provided a set of indicative instances (24 reviews) to serve as the gold standard for reference. We distributed the remaining 1800 reviews equally among 30 scholars. This resulted in 60 annotations done by each scholar to do away with problems of monotony and repetitive behaviors. We asked each of the rest (10 scholars) to monitor the annotations of three scholars and validate them. This exercise resulted in disagreements on 159 reviews, which is less than 10% of the overall set. The first and third authors independently labeled the ambiguous instances, which resulted in a fair agreement with an inter-coder reliability of around 0.51 (Krippendorff, 2008). They discussed and concurred on the labels over two online sessions of 60 min each before reaching a complete consensus. Finally, the entire manual analysis and labeling resulted in 1824 reviews with the following distribution: design-648, defect-433, material-116, and service-627. The detailed sampling process is reported in Table 2.

3.2 | Exploring the constituents—Topic modeling

The central objective of this study is to extract the latent dimensions of quality underlying online consumer reviews. Since we are dealing with a vast corpus of first-hand reviews, the nuances of each review are complicated for manual contemplation; therefore, the latent aspects may remain unobserved (Ding et al., 2020; Fresneda et al., 2021). To address this concern, the study resorts to NLP by using generative models based on the statistical distribution of words. Of all the extant topic modeling techniques (Kumar et al., 2022), structural topic modeling (STM) is one of the most comprehensive techniques for extracting themes from textual corpora (Fresneda et al., 2021; Sharma et al., 2021), with the assumption that words are generated from various probability distributions (Blei et al., 2003; Pang & Lee, 2008). Such underlying latent distributions are inferred as factors or themes comprising a set of the most coherent words (Sharma et al., 2021; Zhan et al., 2018).

Most importantly, by assuming that the latent dimensions can be correlated, STM allows researcher flexibility (Ding et al., 2020), and it helps us to quantify the orientation of each message of self-expression toward all the factors, though in varying degrees. This is analogous to exploratory factor analysis, the most popular approach for dimensionality reduction (Hair et al., 2009). Documents as reviews represent respondents, while words act as the item variables corresponding to the latent factors (Travis et al., 2017). A qualitative judgment of those words within each latent dimension helps assign a context or identity to each factor. A similar analogy exists in other interpretable dimension reduction techniques, such as factor analysis, where item variables are loaded with the latent factors(s) (Kumar & Thakurta, 2021).

3.3 | Identifying the latent characteristics

Our initial exploration revealed that a few of the extracted latent dimensions could not convey a clear notion of the hidden characteristics. Therefore, we resorted to several practitioners with expertise in quality management, grievance redress, and service delivery mechanisms to attach more relevance to the hidden themes. We could seek consent from eight practitioners willing to help us with the initial theme identification process. Unbiased opinion warrants independent analysis; accordingly, four independent groups were formed

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and mapped to the four sources based on their experience and dominion. Each group consisted of two practitioners, of which at least one had significant experience in e-commerce (the Appendix in the Supporting Information). The preliminary investigation resulted in several points of disagreement within each panel. The first author initiated a joint meeting for all four evaluation panels to construe the reasons behind the disagreement. After this intervention, the second author was consulted for his remarks in the final round to uncover coherent and meaningful themes, finally observed as characteristics underlying four sources of dissatisfaction.

3.4 | Empirical investigation of interrelationships (partial least squares based structural equation modeling [PLS-SEM])

We perform an econometric investigation to attend to our central inquiry and second research question. Following an empirical evaluation, we attempt to construe the impact of the four QISCD and their inherent characteristics on future recommendation behavior. We focus on customer responses regarding their ability to influence or discourage future purchase. Based on the theoretical underpinnings of ER and ECT, we also investigate the intermediary role of emotional cognizance and writing style. We use R as an open-source platform to capture sentiments, emotions, and authorial features through their built-in packages—Syuzhet, Quanteda, and SentimentAnalysis—to capture pertinent features (Williams, 2016). While uncovering the nuances of sentiments and emotions, we remain consistent with the extant literature. Since our study delves deep into customer dissatisfaction, we only capture the negative sentiments, which is the frequency of terms classified as representing negative sentiments in the NRC word Lexicon, popularly known as EmoLex.[9] Essentially, each word of the review is matched with the lexicons of QDAP[10] dictionary to assign scores for the negative words, and later these scores were aggregated at a review level (Feuerriegel & Pröllochs, 2021). Further, we compute the eight basic emotions from the psychology literature: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust (Chatterjee, 2019; Kumar et al., 2022).

In this study, we conduct a thorough examination of the authorial characteristics present in reviews, aiming to understand the influence of QISCD on writing style and the extent to which this writing style affects product discouragement. Building upon previous research, we concentrate on the verbosity or succinctness of reviews by analyzing sentence and word length, the frequency of content (meaningful) words, and the presence of stop (function) words to evaluate their overall meaningfulness (Kumar et al., 2023a). Prior studies, such as those by Banerjee and Chua (2017), have demonstrated that reviews containing a higher proportion of function words such as articles, prepositions, and parts of speech tend to be perceived as less coherent. Additionally, we explore the use of punctuation and symbols to determine the degree of casualness or seriousness in expressions of dissatisfaction (Kumar et al., 2023b). Consequently, our analysis encompasses item variables such as sentence count, word count, stop-word count, symbols, total tokens, punctuation, and numbers.

The results of STM provide us the means to attach orientation to each review toward the explored latent characteristic. This helps us generate a matrix of independent responses to capture their influence on future recommendations. Since our conceptualization envisages a myriad of interrelationships, using PLS-SEM seems apt. as In the past, PLS-SEM has demonstrated supremacy over covariance-based structural equation modeling and simultaneous multiple regression models due to their distributional properties, such as multivariate normality and factor indeterminacy, including measurement errors as composites against factors (Akter et al., 2016; Hair et al., 2009).

Essentially, PLS-SEM has gained prominence for analyzing multiple constructs, with several mediation effects. In this regard, recent works underscore the nuanced understanding of indirect effects and in establishing the significance of intermediary relationships (Sarstedt & Moisescu, 2024). Moreover, the use of PLS-SEM has complemented computational intelligence approaches like NLP and machine learning to yield profound theoretical insights. Therefore, PLS-SEM has successfully demonstrated its efficacy in business research, while catering to modern analytical demands (Richter & Tudoran, 2024). Given the complex manifestations of consumer psyche and decision-making, this method has proven effective in modeling complex relationships (Kumar et al., 2023a). PLS-SEM is considered as one the most equipped methods to analyze sequential mediation effects in varied practical settings (Kumar et al., 2025a, 2025b). Most importantly, PLS-SEM allows bootstrapping by performing analysis over a large number of sub-samples (say 1000) and provides robust estimates of path coefficients, loadings, and other model parameters through resampling techniques. Since PLS-SEM presents the standard error and interval estimates of all the path coefficients, it improves reliability, statistical power, and validity of the analysis (Ringle et al., 2012). Due to the above-mentioned reasons, PLS-SEM is our obvious choice to carry out the investigation in two stages: building the measurement model for a reliable scale and a structural model to test the interrelationships (Akter et al., 2016).

4 | Results

It is imperative to identify the number of latent dimensions to explore the QISCD and extract pertinent, meaningful characteristics with minimal overlap of notions. We rely on exclusivity and semantic coherence, the two metrics that attend to our above-stated objective (Ding et al., 2020; Sharma et al., 2021).

[9] https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm.
[10] https://cran.r-project.org/web/packages/SentimentAnalysis/readme/README.html.

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TABLE 3 Sensitivity for extracting the number of dimensions.

	Defect		Design		Material		Service	
K	Exclusivity	Semantic coherence	Exclusivity	Semantic coherence	Exclusivity	Semantic coherence	Exclusivity	Semantic coherence
2	7.138269	−80.97549	7.101208	−89.64156	7.079129	−73.27194	8.023943	−54.66524
3	8.379492	−82.47412	8.49313	−90.03566	7.988694	−84.34603	8.508724	−56.18608
4	8.971855	−87.7217	9.062344	−96.87086	8.477352	−91.62874	9.075299	−63.97831
5	9.217923	−93.8134	9.186956	−97.9164	8.871498	−86.48121	**9.126146**	−59.50159
6	9.244043	−94.32858	9.443695	−106.1943	**8.968583**	−88.60788	9.083515	−66.37302
7	9.471995	−99.19486	9.474102	−107.6846	8.917104	−98.39863	9.297003	−71.19024
8	9.48398	−99. 34324	9.587958	−111.5762	8.995228	−93.98596	9.22682	−70.33509
9	**9.57458**	−99. 99569	9.606264	−111.6935	8.987198	−95.85907	9.312169	−65.92019
10	9.646551	−101.8177	9.633646	−112.1456	9.127978	−94.61883	9.298809	−70.79501
15	9.736824	−108.0851	9.763415	−123.6887	9.302223	−106.491	9.52025	−79.48945
20	9.784911	−114.9071	9.807777	−126.0318	9.454805	−113.2434	9.545506	−85.11676
25	9.838019	−122.1504	9.83802	−129.9017	9.493906	−111.613	9.600418	−86.47521
30	9.857536	−125.753	9.863135	−136.4016	9.566587	−114.2326	9.597291	−87.45689
40	9.889934	−137.4806	9.882335	−142.9577	9.488231	−116.0281	9.655754	−93.60445
50	9.832277	−125.7824	9.813731	−127.4333	9.463523	−118.2566	9.646151	−93.40467

4.1 | Identifying the constituents of QISCD

The exclusivity metric indicates the uniqueness of the terms describing each topic; the semantic coherence metric echoes the unified notion of the latent dimension(s) (Fresneda et al., 2021). The two metrics are the facets of the frequency of co-occurring words corresponding to each topic. With increased dimensions, the hidden content overlaps and shows weak coherence. On the contrary, exclusivity initially increases till it saturates at an interim value. We report both exclusivity and semantic coherence for reviews pertaining to the four sources (Table 3).

For material and service, exclusivity peaks before tapering off, hinting toward an optimal number of dimensions. The semantic coherence—corresponding to six themes for material and five themes for service—is contained. The exclusivity for defect improves at a decreasing rate beyond nine dimensions; accordingly, we make a suboptimal choice with nine latent dimensions underlying the defect. The choice was not so explicit for design, where exclusivity and semantic coherence showed a monotonic improvement. After manually interpreting various dimensions, we finally settled on seven latent themes underlying design.

We document the characteristics of QISCD (Table 4) as per the qualitative assessment (Section 3.3). The reference point for our exploration and identification was the extant scales and instruments within the quality management scholarship (Garvin, 1987; Yu & Fang, 2009). Topic modeling was conducted for each of the four sources to explore and identify the latent characteristics underlying each of the four sources. The STM analysis yields four categories of word distribution on the latent themes, namely: Highest, FREX, LIFT, and Score. While Highest is used to arrange the words in each latent distribution based on their likelihood of occurrence, score is an indication of which words make the theme coherent. While FREX focuses on exclusive words occurring frequently (FREX score) within each dimension. We also accounted for rare and unique words (LIFT score) that were almost absent in the other themes, making our themes exclusive and non-overlapping (Sharma et al., 2021). To clearly represent themes, we selected words with high FREX and LIFT scores, as these metrics help capture both uniqueness and thematic relevance (Kumar et al., 2022, 2023a). Accordingly, the words listed under each category in Table 4 exhibit high values for both FREX and LIFT.

Since most of our constructs are unidimensional, their internal reliability regarding Cronbach's alpha remains 1. As explained earlier (Section 3.1), the final 1824 reviews belonged to one of the four categories: design, defect, material or service, which is denoted as binary presence. For example, a value of one for defect would indicate that the major QISCD is a defect but not the other three categories. Likewise, sentiment is also a unidimensional construct measuring the score of that review signaling negative tone. For other multidimensional constructs, authorial features have Cronbach's alpha of 0.873, and emotional valence has Cronbach's alpha of 0.912. The two constructs, authorial features and emotional valence, have with respective loadings of seven and 10 items listed above (described in Section 3.1). We can state that the items loaded on these two constructs capture the reflections of emotions and writing style (Hair et al., 2009). The convergent validity is 1 for all unidimensional constructs, while 0.746 for authorial features, and 0.659 for emotions. Authorial features and emotions are computed as average variance extracted for their items as a measure for convergent validity (Table 5; Hair et al., 2009; Sarstedt et al., 2014).

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TABLE 4 List of identified characteristics alongside some indicative extracted words underlying the four sources of quality-induced customer dissatisfaction.

Defect	Design	Material	Service
Crack [drip, leak, crack, seal, drain]	Noise [thing, nois, make, fill]	Cheap [return, product, start, cheap, materi, back]	Responsiveness [seller, item, contact, part, appoint, email]
Malfunction [stop, fail, unit, dry]	Leakage [leak, flow, filter, tap]	Weathering [use, product, month, problem, rang, nasti, whole]	Reliability [replace, expire, repair, bad]
Noise [nois, loud, sent, back]	Aesthetics like, look, realli, design, use]	Serviceability [replac, coverag, filter, standard, genuin, part]	Tangibles [servic, warranti, provid, custom, manufactur, number]
Conformance [base, receiv, wrong, order, pictur, descript]	Fit [fit, model, small, size, proper]	Safety [flammabl, test, vent, ignit]	Assurance [protect, week, schedule, ship, order, arriv]]
Missing Parts [miss, part, oem, screw, replac, genuin]	Suitability [work, fail, waste, junk]	Conformance [thinner, intermittent, poor, shorter, tag]	Empathy [told, wait, ask, come]
Durability [stop, work, break, time]	Installation [part, instruct, wrong, instal, pictur]	Availability [elsewhere, origin, buy, part, unavil]	
Perceived Quality [absolut, garbag, wast, money, spend, save, junk]	Reliability [stop, day, work, week, month,]		
Serviceability [year, warranti, repair, new, fix]			
Fit [differ, model, number, correct, fit]			

TABLE 5 Reliability and validity measures of the constructs.

Constructs	Cronbach's alpha	rhoC	rhoA	Average variance extracted
Design	1	1	1	1
Defect	1	1	1	1
Material	1	1	1	1
Service	1	1	1	1
Authorial features	0.873	0.909	0.948	0.746
Negative sentiment	1	1	1	1
Emotions	0.912	0.926	0.925	0.659
Future recommendations	1	1	1	1

Table 6 provides a detailed overview of the statistics for all variables related to the four QISCD. To understand the patterns in these variables, we examine measures of location such as mean, maxima and third quartile (Q3), which offer insights into the distribution of values. Additionally, we assess the spread using the inter-quartile range to understand the dispersion of the middle 50% of values, while skewness highlights any asymmetry in distribution. Given that the distinct characteristics of each QISCD are different, we delve into the detailed findings of inner models in the subsequent section. Starting with authorial features, and it is evident that customers tend to elaborate extensively on service-related issues, as indicated by significant verbosity in word, sentence, and token length. Simple word count and content count denote the presence of meaningful or content-specific words and grammatical words, respectively. When expressing dissatisfaction with design-related issues, the mean proportion of content words is higher, compared to stop words, suggesting greater average meaningfulness in these concerns. Similarly, emotions related to service aspects are generally heightened, while negative sentiments are consistent across all four QISCD.

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TABLE 6 Descriptive statistics highlighting measures of location, spread, and shape for all variables corresponding to the four quality-induced sources of customer dissatisfaction (QISCD).

	Authorial features							Sent score	Emotional valence										Customer dis-satisfaction	Customer response
	Sent count	Word count	Stop count	Tokens	Types	Puncts	Num	*Sent Neg*	Anger	Anti-cipation	Disgust	Fear	Joy	Sad	Surprise	Trust	Neg	Pos	*dissatis score*	*Fut recomm*
Mean																				
Material	4.76	31.81	49.65	7.80	0.79	0.13	0	0.10	0.55	1.13	0.41	0.68	0.62	0.80	0.43	1.29	1.78	1.89	1.32	0.23
Defect	4.51	31.58	47.95	7.785	0.87	0.18	0.003	0.1	0.68	1.046	0.615	0.69	0.65	0.79	0.447	1.19	1.8	1.89	1.27	0.19
Design	3.65	27.33	38.46	6.03	0.66	0.09	0.003	0.10	0.56	0.80	0.49	0.56	0.50	0.67	0.34	0.93	1.42	1.48	1.26	0.20
Service	8.68	43.61	89.05	17.62	2.30	0.46	0.020	0.10	1.23	2.20	1.09	1.54	1.30	1.73	0.86	2.85	3.57	4.48	1.17	0.20
Maximum																				
Material	26	81	217	53	8	4	0	0.56	5	11	3	4	6	4	6	10	9	17	2	1
Defect	30	81	256	94	26	19	1	0.67	6	10	8	7	7	7	6	11	14	15	2	1
Design	74	104	646	282	55	18	2	0.56	10	23	9	13	27	15	7	36	24	72	2	1
Service	100	117	769	309	36	43	7	0.38	17	29	19	34	30	25	15	33	56	63	2	1
Q3																				
Material	5	37	55	8	1	0	0	0.14	1	2	1	1	1	1	1	2	2	3	2	0
Defect	5	35	52	9	1	0	0	0.14	1	2	1	1	1	1	1	2	2	3	2	0
Design	3	25	33	5	0	0	0	0.08	0	0	0	0	0	0	0	1	1	1	1	0
Service	11	54	107	20	3	0	0	0.13	2	3	2	2	2	2	1	4	5	6	1	0
IQR																				
Material	2	12	20	4	1	0	0	0.09	1	2	1	1	1	1	1	2	1	3	1	0
Defect	2	8	15	4	1	0	0	0.09	1	2	1	1	1	1	1	2	1	2	1	0
Design	2	9	12	4	1	0	0	0.1	1	1	1	1	1	1	1	1	2	2	1	0
Service	7	22	60	13	3	0	0	0.07	2	2	2	2	2	2	1	3	4	4	0	0
Skewness																				
Material	3.26	1.284	3.004	3.725	2.58	5.56	_	1.73	1.99	2.866	1.706	1.39	2.27	0.94	3.446	2.01	1.54	2.86	0.77	1.31
Defect	3.13	1.226	3.315	4.023	4.34	12.8	17.75	1.07	1.57	1.702	1.644	1.71	1.71	1.72	1.845	1.9	1.74	1.98	1.05	1.55
Design	7.23	1.951	7.574	15.26	10.6	13.5	21.60	0.86	2.09	3.49	2.082	2.81	5.88	2.88	2.418	5.45	3.23	9.79	1.09	1.48
Service	4.5	0.526	3.039	5.829	3.66	13.6	21.71	1.05	2.83	3.16	3.703	6.8	5.83	3.4	3.124	2.82	4.76	4.42	1.74	1.51

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TABLE 7 Path coefficients and significance of the inner models corresponding to the four QISCD.

	Defect				Design		
	Estimate	***t*-stat**	***Pr*(>|*t*|)**		**Estimate**	***t*-stat**	***Pr*(>|*t*|)**
Intercept	−5.68E−17	−1.34E+00	0.181	Intercept	−2.53E−17	−1.26E+00	0.209
Crack	5.10E−01	1.19E+16	0	Noise	2.19E−01	1.06E+16	0
Malfunction	3.38E−01	7.86E+15	0	Leakage	5.33E−01	−2.41E+16	0
Noise	3.46E−01	8.00E+15	0	Aesthetics	2.81E−01	1.35E+16	0
Conformance	3.62E−01	8.15E+15	0	Fit	3.92E−01	1.90E+16	0
Missing parts	4.63E−01	1.06E+16	0	Suitability	2.91E−01	1.39E+16	0
				Installation	3.93E−01	1.83E+16	0
Durability	3.66E−01	8.45E+15	0	Reliability	2.47E−01	1.22E+16	0
Perceived quality	4.74E−01	1.11E+16	0		Material		
					Estimate	*t*-stat	*Pr*(>|*t*|)
Serviceability	2.77E−01	6.39E+15	0	Intercept	−4.81E−17	−1.54E+00	0.126
Fit	3.92E−01	9.14E+15	0	Cheap	3.86E−01	1.22E+16	0
	Service			Weathering	3.42E−01	1.08E+16	0
	Estimate	t−stat	Pr(>|t|)	Serviceability	6.41E−01	2.03E+16	0
Intercept	4.27E−17	1.25E+00	0.213	Safety	4.02E−01	1.27E+16	0
Responsiveness	5.64E−01	1.60E+16	0	Conformance	5.27E−01	1.67E+16	0
Reliability	2.96E−01	8.26E+15	0	Availability	4.59E−01	1.45E+16	0
Tangibles	4.01E−01	1.13E+16	0				
Assurance	6.05E−01	1.75E+16	0				
Empathy	5.15E−01	1.47E+16	0				

Further analysis of location and shape measures reveals that the high meaningfulness of design-related concerns may stem from a limited number of reviews, as both maximum values and skewness are notably higher, compared to the other QISCDs. This suggests that individuals with a deep understanding of the product can clearly articulate design-related issues, while others may struggle to ascertain the core issues. Interestingly, service-related reviews exhibit varied sentence lengths, reflecting individual orientation toward concise versus detailed expression. Finally, we also observe that the average customer dissatisfaction and tendency to discourage products are higher for materials, compared to the other QISCDs; however, the difference is statistically insignificant. Measures of spread and shape are similar for both customer dissatisfaction and future recommendation.

Gauging the influence of each latent characteristic on the corresponding four QISCD (Table 7), we observe the influence of all the extracted characteristics and notice the presence of cracks concerning any defect, pivotal in shaping customer resentment. Both fitment issues and missing component parts foster defect-induced customer disappointment. The perception of defect is influenced by the unit presence of cracks (0.591 units), an absence of missing parts (0.463 units), and fitment issues (0.392 units).

Leakage is a shortcoming of the entire assembly line for that product. Improper installation is another characteristic of design-related issues. Design-related flaws are significantly reflected by 0.519 units for a unit realization of leakage and 0.393 units from installation problems. Characteristics like noise, installation, and fit can be attributed to both design and defect. Still, problems with design would indicate that the entire product line, not any random product, is faulty. If noise is a concern with much of the production line, the concern may be due to design issues and have little impact (0.219), but if the noise is exclusive to any one unit, the impact is incremental (0.346), almost 1.5 times that of a design issue.

Serviceability—the ability of a specific material to undergo repair, processing, cleaning, and use—is the most influential trait (Garvin, 1987). Conformance is a product's conformity to the manufacturer's guidelines and specifications (Garvin, 1987; Yu & Fang, 2009). If materials are substandard or repair is not possible, materials-related customer resentment will increase by 0.641 units and 0.527 units, respectively. Assurance suggests that the team can build trust among customers and inspire them to feel confident about the resolution. Responsiveness estimates the team's commitment to resolving issues. The role of service quality in e-commerce cannot be overestimated, so assurance and responsiveness result in service-related customer satisfaction. A unit mention of concerns over assurance and responsiveness will influence service-related customer dissatisfaction by 0.605 units and 0.564 units, respectively.

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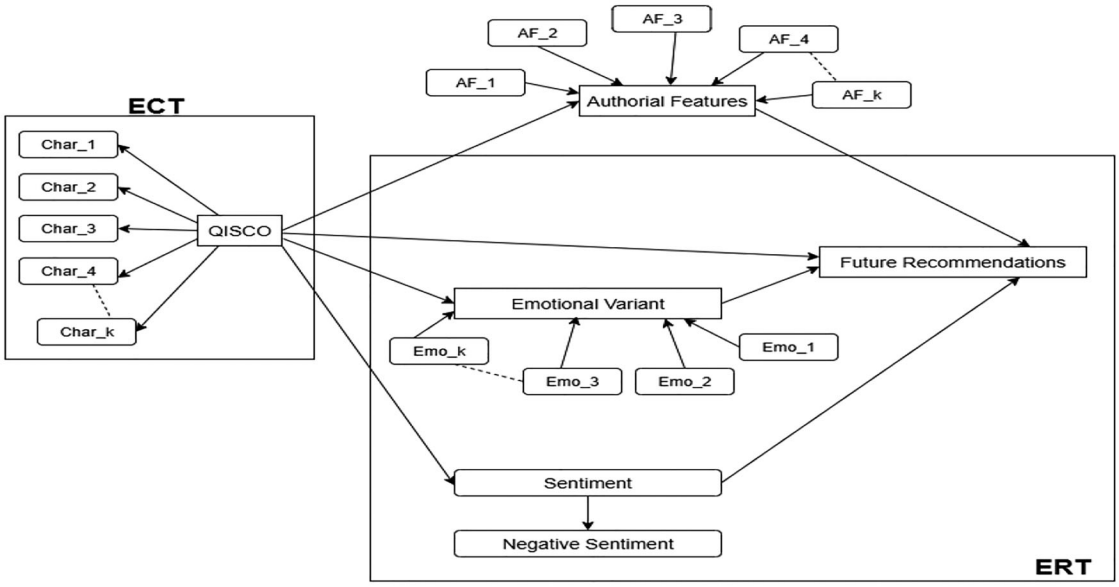


FIGURE 2 Overall conceptual diagram to depict the posited structural relationships on how individual quality-induced sources of customer dissatisfaction may influence negative response behaviors.

4.2 | Empirical model-impact on future recommendation

Word of mouth plays an instrumental role in the success of e-commerce; and QISCD influence customers to make negative recommendations and discourage potential customers. Motivated by the theoretical lens of ECT, our second research question enquires into the influence of the four major QISCD and their inherent characteristics on negative response behaviors. The theory of ER advances our investigation into envisaging the intermediary role of writing style, sentiments, and emotional cognizance spurred from sources of dissatisfaction and resulting in product discouragement.

For consistency and robust findings (Ringle et al., 2012), we report and describe the bootstrapped results, where models have been executed over 1000 iterations. Figure 2 helps us envision the overall conceptual model, while the subsequent tables help in uncovering the nuances of structural paths Tables 8–11. To begin with, we demonstrate the bootstrapped mean and interval estimates of all the structural paths. Only those estimates are statistically significant whose lower and upper interval limits have the same sign; in other words, the interval does not contain zero. The interpretation is that the hypothesized no relation (zero coefficient) is out of the observed interval depicting a significant relation. In Table 8, we can see that defect significantly influences negative sentiments, writing style, and emotional valence of dissatisfied customers. The bootstrapped results for the inner loadings of all the four QISCD are reported in the Appendix in the Supporting Information. Since previously we have exfoliated the factor weights of the inner model, we avoid superfluous discussion about the underlying items of defect. For other multi-dimensional constructs, almost all authorial features are significantly loaded with their corresponding items, while anger, anticipation, sadness and negative emotions amongst are prominent for the emotional valence.

A unit increase in realizing defects and their underlying characteristics will increase negative sentiments by 0.058 units and influence emotions by 0.155 units. Defects and their underlying characteristics also influence writing style by 0.14 units. However, we do not see a significant direct relation between defect and future recommendation, eliminating any possibility of direct effects of defects on negative response behavior. Therefore, we further explore the presence of indirect effects, and we observe that emotions significantly mediate the impact of experiencing defects over purchase discouragement. The significant indirect path alongside non-significant direct paths reflects a full mediation pattern in the triadic relationship of defect, emotions and purchase discouragement. This suggests that the effect of defect on FR operates primarily through its influence on emotions rather than defect directly impacting FR. Based on related literature (Benitez et al., 2020; Zhao et al., 2010), this establishes a full mediation effect of emotions in the impact of defect and its underlying characteristics over purchase discouragement.

We next look into how much of the design and its underlying characteristics impact negative response behaviors. Design and its attributes significantly reduce emotional cognizance and sentiments by 0.2 and 0.132 units, respectively, while increasing purchase discouragement by 0.005 units. Nevertheless, the direct impact of design on purchase discouragement lacks statistical significance. Thus, we now interpret the intermediary relation through indirect effects. The bootstrapped estimates reveal that negative sentiments significantly mediate the impact of design and its underlying

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TABLE 8 Structural paths with direct and indirect effects for defect-related customer discontent.

Indicator	Original est.	Bootstrap mean	Bootstrap SE	2.5% CI	97.5% CI
Defect → AF	0.14	**0.14**	0.069	0.004598439	0.275401561
Defect → Sentiments	0.054	**0.058**	0.029	**0.001092098**	**0.114907902**
Defect → Emotions	0.151	**0.155**	0.078	**0.001937366**	**0.308062634**
Defect → FR	0.009	0.007	0.021	−0.034209171	0.048209171
AF → FR	0.006	0.003	0.025	−0.046058537	0.052058537
Sentiments → FR	−0.007	−0.008	0.016	−0.039397463	0.023397463
Emotions → FR	−0.035	−0.031	0.025	−0.080058537	0.018058537
Defect → AF → FR	0.0009	0.00035	0.0014	−0.002397278	0.003097278
Defect → Emotions → FR	0.0052	**0.00569**	0.0022	**0.001372849**	**0.010007151**
Defect → Sentiments → FR	0.0003	0.00059	0.0008	−0.000979873	0.002159873

TABLE 9 Structural paths with direct and indirect effects for design-related customer discontent.

Indicator	Original est.	Bootstrap mean	Bootstrap SE	2.5% CI	97.5% CI
Design → AF	−0.151	−0.04	0.15	−0.334351219	0.254351219
Design → Sentiments	−0.132	**−0.132**	0.012	**−0.155548098**	**−0.108451902**
Design → Emotions	−0.204	**−0.205**	0.039	**−0.281531317**	**−0.128468683**
Design → FR	0.005	0.005	0.012	−0.018548098	0.028548098
AF → FR	−0.005	−0.002	0.018	−0.037322146	0.033322146
Sentiments → FR	−0.007	−0.007	0.012	−0.030548098	0.016548098
Emotions → FR	0.008	0.008	0.018	−0.027322146	0.043322146
Design → AF → FR	0.00079	0.00087	0.0026	−0.004232088	0.005972088
Design → Emotions → FR	−0.00154	−0.00166	0.0038	−0.009116898	0.005796898
Design → Sentiments → FR	0.000976	**0.0032**	0.0016	**6.02537E-05**	**0.006339746**

TABLE 10 Structural paths with direct and indirect effects for material-related customer discontent.

Indicator	Original est.	Bootstrap mean	Bootstrap SE	2.5% CI	97.5% CI
Material → AF	0.344	0.206	0.401	−0.580898926	0.992898926
Material → Sentiments	0.12	0.088	0.132	−0.171029073	0.347029073
Material → Emotions	0.42	**0.402**	0.196	**0.017381074**	**0.786618926**
Material → FR	0.016	−0.001	0.131	−0.258066731	0.256066731
AF → FR	−0.028	−0.014	0.119	−0.247518634	0.219518634
Sentiments → FR	0.062	0.061	0.071	−0.078326244	0.200326244
Emotions → FR	0.204	0.041	0.237	−0.424074926	0.506074926
Material → AF → FR	−0.0095	−0.0048	0.0522	−0.107234224	0.097634224
Material → Emotions → FR	0.0857	0.0581	0.0954	−0.129107375	0.245307375
Material → Sentiments → FR	0.0074	0.00657	0.0144	−0.021687717	0.034827717

characteristics on future purchase discouragement. Similar to defect, we also observe full mediation for the role of design, resulting in product discouragement through negative sentiments. This highlights that a gush of emotions and negative sentiments carries the burden of unmet expectations up to product discouragement. However, without the intermediary role of negative sentiments, direct effects are insignificant (Zhao et al., 2010). Moving further, we explore the nuances of customer expressions and feelings through the inner model loadings (Appendix B in the Supporting Information) of other multi-dimensional constructs. Mostly, consumer expression is observed as verbose, while using fewer symbols and less

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TABLE 11 Structural paths with direct and indirect effects for service-related customer discontent.

Indicator	Original est.	Bootstrap mean	Bootstrap SE	2.5% CI	97.5% CI
Service → AF	−0.206	−0.182	0.149	−0.474388878	0.110388878
Service → Sentiments	−0.07	−0.047	0.062	−0.168665171	0.074665171
Service → Emotions	−0.212	−0.227	0.167	−0.554711024	0.100711024
Service → FR	−0.069	−0.032	0.056	−0.141891122	0.077891122
AF → FR	−0.051	−0.007	0.073	−0.150250927	0.136250927
Sentiments → FR	−0.024	−0.016	0.037	−0.088606634	0.056606634
Emotions → FR	0.041	0.032	0.068	−0.101439219	0.165439219
Service → AF → FR	0.0105	0.0055	0.0146	−0.023150185	0.034150185
Service → Emotions → FR	−0.00869	−0.0081	0.0178	−0.043029678	0.026829678
Service → Sentiments → FR	0.00164	0.00101	0.00309	−0.005053635	0.007073635

punctuation. Moreover, anger, sadness, trust issues and negative emotions dominate the emotional valence with high and significant loadings. Tables 9–11

Materials and their attributes do not reflect any significant signs of product discouragement; however, we see that material and its underlying characteristics can influence emotions, negative sentiments, and writing style, by 0.206, 0.088, and 0.4023 units, respectively. From all, we can interpret that materials-related issues significantly stimulate emotional cognizance, which propels negative recommendation behavior. However, we fail to establish any significant indirect effect of design on future purchase discouragement. For other multi-dimensional constructs, we see that authorial features are denoted through verbosity, function words, use of symbols and punctuations, reflecting a disparate focus in writing style. While sadness and negative emotions are the major underlying emotions.

Finally, we proceed to understand similar interrelationships for service. Service and its underlying attributes do not influence product discouragement. All the relationships indicate that service-related concerns do not influence product discouragement directly or indirectly. Last, to demonstrate the impact of the multidimensional notion of the four sources of quality-induced customer dissatisfaction, we state the findings for our first hypothesis: neither of the four stand-alone sources shows evidence of product discouragement. Even the mediating effects of emotions and sentiments are statistically insignificant. Our results show that the genesis of product discouragement is not explained by the mere presence of the four QISCD. Therefore, we conclude that identifying the characteristics of the quality-induced sources of dissatisfaction is imperative.

5 | Discussion

The descriptive summary of various constructs unveiling writing style, tonal qualities, and emotional gush reveals that dissatisfied customers elaborate more extensively on service issues, indicating that service failures evoke detailed narratives. As feedback for portal managers, the service-related grievances have the potential to shape future redressal mechanisms. Emotional intensity is notably higher in service-related reviews, indicating that service-related sources are having a profound impact on consumer psyche. Design-related complaints, characterized by a higher proportion of content-specific words and greater skewness, suggest that only a technically informed subset can clearly articulate such flaws. If given due importance, such reviews could be instrumental for product manufactures in stimulating design considerations. The variability in sentence length reflects different customer communication styles, highlighting the need to avoid oversimplified conclusions based solely on review length. Furthermore, material-related dissatisfaction shows a stronger tendency to drive product discouragement, highlighting its potential to amplify negative word-of-mouth and impact brand perception across all QISCD.

We could not gather enough evidence to support our first hypothesis, so we concluded that none of the four broad QISCD can explain future recommendation behaviors. This study then deep-dives to unveil the constituents of the four multidimensional sources of dissatisfaction and see how the nuances can explain purchase discouragement over online portals. The identified constituents are several attributes involving traits related to product and service quality from the quality management scholarship. However, the direct or indirect influence of a service and its inherent attributes on future recommendations is not statistically significant. The theory of ER could most likely explain the reason. Service-related flaws would probably result from previous complaints about products or related transactions. Since service-related concerns may occur at the second stage of the consumption process, these have a limited negative impact on word-of-mouth.

Next, we delve into other QISCD, particularly defect and design, which share similar inherent attributes such as noise and fit. Our results indicate differences in the magnitudes and importance of these common constituents for the two sources. A product may be noisier due to a defect, but if a whole lot of production echoes noise, the cause is a design fallacy. Design

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fallacies may not be considered important since the concern pertains to the product development phase rather than defects, resulting from a deficient production process. Likewise, noise surfaces as a severe attribute for product discouragement when underlying a design problem. However, fit is equally problematic, whether pertaining to a defect or a design issue. The serviceability of the entire unit has less impact than the serviceability of the material(s) used in the product; therefore, it is more important that parts and their materials can be refurbished and reprocessed rather than repaired.

We also explore the intermediary role of emotions, sentiments, and even writing style on QISCD and future recommendations. Our findings align with the underpinnings of the theory of ER and the ECT. Customer evaluation alone may not directly influence negative recommendations, but it may stimulate negative emotions and sentiments, and psychological cognizance can result in product discouragement. Similar reflections are observed for writing style. Consumers experience significant emotional cognizance for defect-related concerns, and they tend to react through symbols and unusual expressions. This establishes how disappointment stemming from and its underlying characteristics triggers negative emotions due to unmet expectations, which ultimately fosters negative response behaviors in the consumer's mind. Likewise, we observe a significant intermediary role of negative sentiments for design-related issues and see how negative sentiments indirectly lead to purchase discouragement. Such findings enable us to further substantiate the complementary role of ECT and the theory of ER in explaining the psychological manifestation of customers.

We conclude that most of the QISCD and their inherent characteristics are responsible for stimulating emotional cognizance and negative sentiments, which are responsible for the spread of negative word-of-mouth and the discouragement of future purchase.

6 | Theoretical and managerial implications

While ECT explains the root cause of poor customer experiences and subsequent realization of unmet experiences, ER, on the other hand, informs how customers express their emotional surges by writing negative reviews and discouraging future purchases. Together, ECT and ER theory help us understand the entire landscape of negative response behavior for dissatisfied customers. Our findings about the intermediary role of emotions and sentiments shift the focus from defect or design inconsistencies as an isolated operational issue to the nuanced affective processes that mediate their impact on future behavioral intentions. Rather than assuming that defects directly diminish customers' willingness to recommend, our results suggest that it is the negative feelings elicited by such defects and/or design, which aggravates from mild dissatisfaction to strong frustration, resulting in extreme reactions like product discouragement. Therefore, portal managers should plan coping strategies aimed at alleviating the negative sentiments and unpleasant emotions brewing in the customer's mind.

The second takeaway for academia is that no discipline dealing with the consumer psyche can ignore human cognitive aspects. Therefore, studies on consumer transactions, experiences, perceptions, or response behaviors should draw their motivation from psychological theories. The evidence we have gathered lets us infer that such cognitive psychic expressions are instrumental in maintaining a satisfied and loyal consumer base; therefore, interactions with customers and corresponding service aspects should aim to stimulate personal touchpoints to prevent damage from negative word-of-mouth. Such measures will not erase negative experiences or feelings but will help reduce customer resentment.

Next, we inform the practice, particularly online portals, about the importance of product information, specifications, and the display of prototype and product images. Any deviation from the stated specifications and product details incites negative sentiment and emotion and ultimately leads to product discouragement. Perceived differences between claimed benefits and customer experience result in severe negative recommendations and undesirable word of mouth. So, conformance is vital for both products and constituent materials.

Product design is important because the impact of QISCD related to materials is severe. Any glitch due to product shortcomings impacts consumption-related experience and attitude toward future usage. Manufacturers should pay utmost attention to assurance, reliability, and conformance while developing products; therefore, platform managers should establish and follow quality management and certification standards. Identifying the QISCD enables e-online businesses to enhance product quality, operational efficiency, customer satisfaction, and market competitiveness, thus fostering long-term success. Moreover, instituting pre-emptive protocols and measures would limit the spread of negative word-of-mouth and prevent loss of future business potential.

7 | Conclusion, limitations, and future directions

QISCD and their constituents influence future recommendation behavior. This study attempts to determine the prominent sources and, based on practitioner insight, the constituents of those sources. With the intermediary role of emotional cognizance (for defect) and negative sentiments (for design), our study offers profound perspectives and implications. Our inquiry, empirically investigated to test the posited hypotheses, produced several takeaways for the practice. When shared with practitioners, our approach and outcomes received positive recognition. Technology and analytics industry leaders commented that our approach and redress mechanism would reduce time and effort significantly, and e-commerce consultants said that the approach has immense scope for automating routine grievance redress and mitigation processes.

Nevertheless, our study has limitations. It is silent on precipitating intentions or attitude-fostering behaviors. Although our theoretical basis paves the way for capturing

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such manifestations, we alienate to reflect such attitudinal nuances. The study could have identified other sources of dissatisfaction; however, the Amazon reviews dataset failed to reveal additional precursors, and the study is restricted to exploring and uncovering the impact of only the four QISCD and their inherent characteristics. Other e-commerce portals may earmark additional sources, and future studies may adopt our conceptualization to gather fresh perspectives.

The quality management scholarship may have contentions over our insights since the existing instruments were not borrowed directly from the focused works. Future studies may capture and identify the underlying traits purely through an academic exercise and based on extant scales. This might help advance the prominence of Garvin's product-related attributes or the sanctity of the SERVQUAL model. A limitation of our approach is that each review is assigned to only one QISCD category—defect, design, material, or service—capturing just a single source of dissatisfaction. While this simplifies analysis, it may overlook complex cases where multiple issues co-exist. Future research could explore more nuanced modeling techniques to capture overlapping dissatisfaction dimensions within the same review, offering a more comprehensive understanding of consumer discontent.

Overall, this study contributes to the literature on consumer psychology, quality management, and e-tailing, with a myriad of interdisciplinary perspectives, and these contributions can foster effective grievance redress mechanisms and pre-empt customer resentment.

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